# THE OPEN PROBLEMS OF GROTHENDIECK'S 1973 BUFFALO LECTURES, AND WHAT BECAME OF THEM

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Abstract. In the spring and summer of 1973 Grothendieck gave three courses at the State University of New York at Buffalo, on topos theory, on algebraic geometry and on group schemes, whose recordings are online. In them he asks some twenty questions he cannot answer, states conjectures, and hands problems to the audience. We list them from the recordings, with the second at which each is asked, and say what the literature has made of each. Most are settled. Several went against what he expected: his characterisation of presheaf topoi, whose two conditions instead characterise the essential subtopoi of presheaf topoi, the non-existence of a right adjoint $f^!$, the colimits of accessible categories, the descent of immersions that are not quasi-compact, and his doubt that quasi-coherent sheaves can be pulled back fully faithfully along universal effective epimorphisms. Perrin proved his main conjecture on group schemes in 1976, and the effectivity of finite equivalence relations had been answered by Venken in 1971, two years before he asked it. Some answers are ours and are proved in notes that accompany the article. What still looks open concerns large classes of limits in his « problem three », one question that depends on set theory, and descent along non-quasi-compact universal effective epimorphisms in ZFC.

## 1. The lectures and the method

The recordings are on a YouTube channel under the pseudonym « Erwin Schulhoff »: four videos for the topos course (April–July 1973), four for algebraic geometry (May–July) and two for group schemes (June–July), the last two also carrying some sessions of the algebraic geometry course. The edition has made an index of results for each course, with the second at which each statement is made, and reconstructed the blackboards [32]. For this note we searched the transcripts for the places where he says that something is not known, that he cannot prove it, that he expects or suspects it, or where he calls it a conjecture or a problem; we kept the ones he frames as questions, and dropped asides and routine omissions. Each problem below gives the video and the time (a link to that second), how he frames it, and its status, with the references we checked. Where we give an argument of our own, not found in the literature, it is marked $^{(e)}$.

The transcripts are first passes from captions and Whisper that no one has checked, and a few exchanges with the room are hard to attribute. He also corrected himself: on 10 July he retracted several claims of 2 July on algebraic theories (the transitivity of strict cogeneration, a two-step construction, a criterion by strict cogenerators), recorded in the edition's list of corrections [8]. We report the question as finally posed.

## 2. Topoi

Video 2 of the topos course is `PqCQM3Pcl6Q` unless stated.

**Problem 1** (1:26:22 and 7:31:23)**.** *Are the presheaf topoi $\widehat{C}$ exactly the topoi that share all the exactness properties of* **Set***, including those involving infinite limits: arbitrary sums and products distributive, and arbitrary products of epimorphisms epimorphic?*

*Date*: 7 November 2026.

2020 *Mathematics Subject Classification.* 01A60; 18F10, 18C35, 14A15, 14L15.

*Key words and phrases.* Grothendieck, Buffalo 1973, topoi, accessible categories, algebraic theories, fpqc descent, group schemes, open problems.

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He suspects it, says it should not be too hard, and returns to it later as a characterisation of what he calls essential topoi, at least when they have generators.

*Status.* False, and the class his two conditions define can be described exactly[(e)]. Write (A) for « arbitrary products of epimorphisms are epimorphisms » and (B) for « arbitrary products distribute over arbitrary sums ». For a Grothendieck topos $E$, (A) and (B) together are equivalent to each of: $E$ is an essential subtopos of a presheaf topos; for a small generating subcategory with its induced topology, covering sieves are closed under arbitrary intersections; $E$ is totally distributive in the sense of Rosebrugh and Wood. The last equivalence is Lucyshyn-Wright's theorem [21, 1.5.5]; the others are proved in the notes accompanying this article [27], using the axiom of choice. Condition (B) alone says that $E$ is locally connected, and the two conditions are independent: sheaves on the Cantor set satisfy (A) and not (B), sheaves on $\mathbf{R}$ satisfy (B) and not (A). A topos with both and not a presheaf topos is the topos of sheaves on $[0, 1)$ whose open sets are the intervals $[0, a)$: its frame is the complete chain $[0, 1]$, which is completely distributive, so (A) and (B) hold, but it has no nonzero projective object, while a presheaf topos is generated by its projective representables [5]. In the localic case, $\mathrm{Sh}(L)$ satisfies (A) and (B) exactly when $L$ is completely distributive, and is a presheaf topos exactly when $L$ is join-generated by completely join-prime elements. His « essential » is not the essential geometric morphism of later usage; that the class he named turns out to be that of the essential subtopoi of presheaf topoi looks like a coincidence of words. We do not claim that these equivalences are new, only that we have not found them stated.

**Problem 2** (4:05:50)**.** *For a morphism of topoi $f$, can $f_*$ have a right adjoint $f^!$ on sheaves of sets without $f$ being an equivalence?*

He has « a feeling » that it cannot, and concludes that the notation $f^!$ has no reason to exist for topoi.

*Status.* False, even for an open dense immersion, the case he was testing. For the open point $j$ of the Sierpiński space $S$, whose sheaves are the maps $A \to B$ (sections over $S$, then over the open point), $j_*$ is the diagonal $A \mapsto (A \to A)$, which is the constant-sheaf functor of $S$, and its right adjoint is the global-sections functor $\Gamma^{(e)}$. More generally, for a small category $C$ with a terminal object the global-sections functor $\Gamma\colon \widehat{C} \to \mathbf{Set}$, the direct image of $\widehat{C} \to \mathbf{Set}$, has a right adjoint: $\widehat{C}$ is a local topos, and such morphisms are the local maps studied by Johnstone and Moerdijk [17].

**Problem 3** (5:18:59)**.** *Every type of algebraic structure, in a broad sense, should have a classifying topos with a universal model, added axioms should correspond to subtopoi, and the homotopy type of the classifying topos should be a classifying space of the structure.*

He states it as a programme and says he has checked it on a dozen or two examples.

*Status.* Carried out for geometric theories: every geometric theory has a classifying topos with a generic model (Joyal–Reyes; Makkai–Reyes [23]; Hakim for rings and local rings [14]), quotient theories correspond to subtopoi (Caramello [6]), and for groups and categories the classifying topos has the weak homotopy type of the classifying space (Moerdijk [26]). The restriction is that « structure » means geometric theory.

**Problem 4** (8:38:47, 8:46:26)**.** *Can every exactness property be written as « every $D$-functor $I \to C$ factors through a given $D$-functor $I \to I'$ »? Is « every semigroup object is a monoid » an exactness property?*

This comes from a discussion with the room after the session, and who says what is not always clear.

*Status.* Formalised by Jacqmin and Janelidze as exactness sequents of sketches, which cover what they call first-order exactness properties, among them properties of internal structures such as an internal monoid being a group, and not higher-order ones such as having enough

projectives [15]. « Every semigroup object is a monoid » holds only in the trivial category: the left-zero multiplication $X \times X \to X$ is associative, and a unit $e\colon 1 \to X$ for it forces $\mathrm{id}_X = e \circ !$, so $X \cong 1$; since every object carries this multiplication, the category is trivial$^{(e)}$.

Two further questions are answered on the spot. Asked for a criterion characterising sober spaces (1:55:37), he says he knows none that can be read off the topos, since a space and its sobrification have the same sheaves; the intrinsic criteria are standard [18]. And the universal problem for torsors under a group $G$ of a topos $S$ (`ORG14LNb5gE`, 3:15:58) he solves at once with the topos of $G$-objects of $S$, as in Giraud [13].

## 3. Algebraic theories relative to a class of limits

For a class $D$ of limits and colimits, he considers the 2-category $\mathbf{Cat}_D$ of categories with $D$-limits and $D$-functors, and calls a theory algebraic if it is obtained from a set $I$ of base objects by stepwise rules: the free $D$-envelope, adjoining arrows, imposing axioms.

**Problem 5** (8:34:57)**.** *Are the algebraic theories relative to $D$ exactly the theories $C \mapsto \mathrm{Hom}_D(R, C)$ 2-represented by objects $R$ of $\mathbf{Cat}_D$?*

*Status.* Settled: yes for small $R$, which is his own argument and the modern presentation of small categories with limits by generators and relations (sketches, essentially algebraic theories [2, ch. 3]); no in general. For $D$ the small limits, $R = \mathbf{Set}$ is representable, with $\mathbf{Set}^{\mathrm{op}}$ as its category of models in $\mathbf{Set}$, which is not accessible, while stepwise theories have locally presentable categories of models; he gives this example himself on 10 July. For $D$ all small limits the algebraic $R$ are, up to duality, the opposites of locally presentable categories, the form Gabriel–Ulmer duality gives [12].

**Problem 6** (10:03:06)**.** *For which $D$ are the envelope problem (the free $D$-completion of a category) and the localisation problem in $\mathbf{Cat}_D$ solvable? He knows them for finite limits and for topoi, and not for additive, abelian or Karoubian categories.*

*Status.* Settled for every small class $D$ by Kelly's transfinite constructions and two-dimensional monad theory [20, 4], with explicit answers: the idempotent completion for Karoubian categories, the biproduct completion of the free preadditive category for additive ones, the free abelian category of Freyd and Adelman [11, 1], the exact completion of Carboni and Celia Magno [7]. For a large class, such as all small limits, the answers depend on size, as he noted on 10 July: localisations of large categories need not be locally small.

**Problem 7** (`4gynb9uTEE4`, 2:03:54 and 2:10:23)**.** *« Problem three »: is $I \to R$ basic if and only if every strictly full subcategory of $R$ stable under $D$-limits and containing the image of $I$ is $R$ itself? And, conjecturally, can $I$ be enlarged so that no arrows remain to be adjoined, only axioms?*

*Status.* Settled, except for large classes $D^{(e)}$. Write (G) for the condition of the problem; it holds exactly when the transfinite closure of $I$ under $D$-limits exhausts $R$, after $\rho$ steps when the shapes of $D$ have fewer than $\rho$ arrows [19, §3.5]. Assuming, as he does, that envelopes, localisations and the other constructions exist in $\mathbf{Cat}_D$, the notes accompanying this article prove that basic implies (G) for every $D$, and that $I \to R$ is basic if and only if (G) holds and $R$ is algebraic over some base [27]; one coherence step is only sketched. For a small class $D$ every $R$ satisfying (G) is essentially small, hence algebraic, so the equivalence holds with the base fixed. For $D$ all small limits it fails: every set is an equaliser of two maps between powers of 2, so $\{2\} \to \mathbf{Set}$ satisfies (G) after two steps, but $\mathbf{Set}$ is not algebraic, since 2 is not presentable in $\mathbf{Set}^{\mathrm{op}}$ [27]. The second question has a positive answer in the locally presentable case by Gabriel–Ulmer: taking for $I$ the opposite of the $\lambda$-presentable objects, the category of models is the category of $\lambda$-continuous functors on $I$, so that only axioms remain to be imposed [12]. What remains open is the converse for large classes beyond these cases, such as finite limits with all colimits but no exactness condition, and whether the number of closure steps is bounded by a small ordinal when the shapes of $D$ are unbounded. Two

related statements of the course on topoi (`ORG14LNb5gE`), that every object of a later stage is covered by images of earlier ones and that the finite products of a basic $I$ generate, are false as stated: adjoining an arrow $1 \to 1 \sqcup 1$ to **Set** gives **Set** $\times$ **Set**, in which $(1, 0)$ is not covered by the diagonal [27].

Around these he asks whether faithfulness of $\mathrm{Hom}_D(R, C) \to C^I$ for all $C$ suffices for algebraicity (9:40:36, 10:00:19). It does not, as he reports himself later, citing an example he attributes to Isbell (the name is uncertain in the recording; 2:03:54). In the session itself he and the room arrive at lattices, and a large ordinal; **Set** with $I = \{2\}$ is another counterexample, faithfulness holding because every set is a limit of powers of $2^{(e)}$. His hope of a bound on the number of limit operations (8:08:14) he retracted himself on 10 July.

## 4. Accessible categories

**Problem 8** (`4gynb9uTEE4`, 1:31:03)**.** *Is a small 2-colimit in* **Cat** *of accessible categories accessible?*

*Status.* No in general$^{(e)}$. Let $B$ be the free category on arrows $a\colon u \to v$, $b\colon v \to w$, $h\colon u \to w$, map it to the chain $0 < 1 < 2$ and to the free monoid on one generator $t$, sending $a, b, h$ to $t$. The pushout, strict or pseudo, is the one-object category with endomorphisms $\{1, t\}$ and $t = t^2$, since $h = ba$ in the chain; the idempotent $t$ does not split, while accessible categories split idempotents. The three categories are small and idempotent-complete, hence accessible, and the functors are accessible. He expected the answer yes, while noting that a sum of two categories with colimits has none. Directed colimits of accessible categories along accessible full embeddings are accessible (Paré–Rosický [30]). The limit side, which he recalls from SGA 4, is right in modern form [22, ch. 5].

**Problem 9** (`4gynb9uTEE4`, 1:51:31)**.** *Is a full subcategory of an accessible category, stable under limits and with a left adjoint inclusion, accessible?*

*Status.* Yes under an extra hypothesis; in ZFC alone we have not found it settled. When the inclusion is also accessible, the condition he proposed adding, the answer is yes in ZFC: for a locally presentable ambient category this is [2, 2.78]. Under Vopěnka's principle the answer is yes for a locally presentable ambient category [2, ch. 6]. Under the weak Vopěnka principle every limit-closed full subcategory of a locally presentable category is reflective, and this fails in the category of graphs when the weak principle fails [3], [2, 6.22]; Wilson showed that the weak principle does not imply the full one [35]. A reflective, limit-closed, non-accessible subcategory in a model of the weak principle without the full one would show that ZFC does not decide the question; we have not found one in print.

**Problem 10** (`4gynb9uTEE4`, 4:48:17)**.** *If $E$ and $E^{\mathrm{op}}$ are both accessible, is $E$ equivalent to the point?*

*Status.* No: the two-element chain is a counterexample, being, like its opposite, small and idempotent-complete. With small colimits in $E$, as he assumes for his main theorem, $E$ and $E^{\mathrm{op}}$ are both locally presentable, and then $E$ is a preorder, hence a complete lattice (Gabriel–Ulmer [12, Satz 7.13]).

## 5. Algebraic geometry and group schemes

**Problem 11** (`vuys5SNs2eQ`, 3:31:11; see also 2:59:31)**.** *Do immersions that are not quasi-compact descend along a faithfully flat quasi-compact morphism? At 2:59:31: is descent of subschemes exact without retrocompactness? (« you asked for an open problem »).*

He believes the first but cannot prove it; his attempt by a filtered union of quasi-compact opens fails on his own example, the closed points of the line.

*Status.* Both are false. Open, closed and quasi-compact immersions descend along fpqc coverings, and immersions along fppf ones [34, 02L3, 02L6, 02L8, 02YM]; on the underlying sets, locally closed and retrocompact subsets descend along fpqc morphisms, and locally closed ones along fppf morphisms [33, Thms 4.1, 4.2]. The disjoint union of the $\operatorname{Spec}\mathbf{F}_p$ mapped to $\operatorname{Spec}\mathbf{Z}$ is not an immersion, but becomes one after the faithfully flat base change $\operatorname{Spec}\prod_p \mathbf{Z}_{(p)} \to \operatorname{Spec}\mathbf{Z}$; the same subscheme answers the second question. The example, base change included ($\prod_q \mathbf{Z}_{(q)}/p \cong \mathbf{F}_p$), is in a comment of August 2026 on Stacks tag 02YL, which we checked; it is not refereed, and we found no earlier published counterexample. The subscheme is the one he used in class, the closed points, taken after base change.

**Problem 12** (`3ULFIl_2Wis`, 5:04:48)**.** *« The main conjecture »: a connected separable group scheme over a field is an extension of a group of finite type by an affine group, the quotient map faithfully flat and quasi-compact; equivalently a filtered limit of groups of finite type with affine transition maps.*

*Status.* Proved by Perrin [31], a summary of his Orsay work: every quasi-compact group scheme over a field is a filtered limit of algebraic quotients with affine kernels, with transition maps affine from some index on; he notes that the statement had been conjectured in EGA IV. A connected group scheme over a field is quasi-compact, as Grothendieck recalls just after the statement. Perrin's counterexample with abelian varieties of growing dimension is the phenomenon Grothendieck wanted to exclude at 5:11:21, the dimensions of the abelian quotients being bounded.

Asked for a characterisation of schemes among algebraic spaces (6:39:12, a session of the algebraic geometry course recorded on this video), he answers that it would solve the representability problem, and no; there are sufficient criteria and partial results [34, 03XX, 0ADD, 03JI, 0D2W]. Smaller questions are settled. In the algebraic geometry course: whether a Noetherian spectrum, the ascending chain condition on radical ideals, is the same as the ascending chain condition on primes (`51wmmpHand0`, 7:42:27; no: an infinite product of fields has only maximal primes and an infinite, non-Noetherian spectrum$^{(e)}$; see Ohm and Pendleton for rings with Noetherian spectrum [28]); and whether a scheme can have no closed point (`Oqhr-Gt66es`, 0:10:49; yes [34, 01IY]). There he also states that the local schemes are those with a unique closed point. That holds for quasi-compact schemes, but not in general: $W \sqcup \operatorname{Spec} k$, with $W$ a scheme without closed point, has a unique closed point and is not local$^{(e)}$. In the group-schemes course: whether a monoid scheme with a closed unit section is separated (`3ULFIl_2Wis`, 3:56:26; no: adjoin a unit, as an extra rational point, to a non-separated scheme with a rational point $c$ and the constant multiplication $xy = c^{(e)}$).

**Problem 13** (`3ULFIl_2Wis`, 6:26:20–6:27:24)**.** *Along a morphism of universal one-descent, that is a universal effective epimorphism of schemes, do quasi-coherent sheaves descend? Is pulling them back at least fully faithful?*

A listener asks why the topology is not defined by all such morphisms; he answers that it would be the finest one for which representable functors are sheaves, but that these morphisms are hard to handle, that one does not know whether quasi-coherent sheaves descend along them or whether pulling back is even faithful, and that he doubts this can be proved. This is an algebraic geometry session recorded on the group-schemes video.

*Status.* Mostly settled$^{(e)}$, and partly against his doubt [27]. Pulling back quasi-coherent sheaves along any universal effective epimorphism is fully faithful, with no quasi-compactness: test against the square-zero extension $\mathcal{O}_Y \oplus F$ and the affine line. Along a quasi-compact one, descent is effective; this is stated by Mesablishvili [24], and the notes give an independent proof through a finite affine cover, which reduces to the affine case, where universal effective epimorphisms are the pure ring maps and descent holds exactly along pure maps [34, 0EUA, 08XA], [25]. Without quasi-compactness, effectivity can fail: if some nonprincipal ultrafilter $U$ on the primes is generated by $\aleph_1$ sets, for instance under the continuum hypothesis, the

family $\{\operatorname{Spec}\mathbf{Z}_A \to \operatorname{Spec}\mathbf{Z}\}_{A\in U}$ of the Stacks Project [34, 0EUG] is a universal effective epimorphism along which some descent datum, an extension of $\mathcal{O}$ by $\bigoplus_p \mathbf{Z}/p$, is not effective; the obstruction is a clopen set of $\beta\mathbf{N}\setminus\mathbf{N}$ minus the point $U$ that does not extend, which exists at points of character $\aleph_1$ by the butterfly lemma of Fine and Gillman [29]. In ZFC alone the question is open: in the model of van Douwen, Kunen and van Mill the obstruction used here vanishes.

## 6. Effectivity

At the end of the algebraic geometry session of 11 July, recorded on the second group-schemes video (`3ULFIl_2Wis`, 6:59:50), asked for open problems, he gives the effectivity of finite equivalence relations on a finite scheme over a field, posed in FGA in 1961, and says he suspects a negative answer. It had been answered negatively, over every field, by Venken in 1971: Corollary 3.1 of a note on the descent of flat modules gives a non-effective equivalence relation on the spectrum of a local Artin algebra of dimension 33 [9]. A companion note checks that example by machine and gives a second one, of length 15, by restricting Raicu's relation on the plane [10].

## 7. Summary

| Problem | His stance | Status |
|---|---|---|
| 1 presheaf topoi | suspects yes | false: Scott sheaves on $[0,1]^{(e)}$; essential subtopoi of $\widehat{C}$ [21] |
| 2 $f^!$ | feels no | false; local morphisms [17] |
| 3 classifying topoi | programme | done for geometric theories |
| 4 exactness properties | open in discussion | formalised [15]; semigroup case trivial$^{(e)}$ |
| 5 algebraic = representable | aim | yes for small $R$, no in general |
| 6 envelopes, localisations | known in two cases | small $D$: yes [20, 4] |
| 7 problem three | neither direction proved | $\Rightarrow$ yes; $\Leftrightarrow$ for small $D$; fails for all limits$^{(e)}$ |
| 8 colimits of accessible | expects yes | no$^{(e)}$; directed, full embeddings: yes [30] |
| 9 reflective subcategories | asks for a counterexample | yes if the inclusion is accessible, or under Vopěnka [2]; ZFC alone open to us |
| 10 $E$, $E^{\mathrm{op}}$ accessible | guesses the point | no: the two-element chain [12] |
| 11 descent of immersions | believes yes | no (Stacks comment, 2026) |
| 12 structure of group schemes | main conjecture | Perrin 1976 [31] |
| 13 descent of quasi-coherent sheaves | doubts it | fully faithful; effective if quasi-compact; can fail under CH$^{(e)}$ |
| §6 effectivity | suspects no | no, over every field (Venken 1971) [9, 10] |

The questions that still look open are the converse of problem three for large classes $D$ and the bound on its number of steps (Problem 7), the status in ZFC alone of Problem 9, and the effectivity of descent of quasi-coherent sheaves along non-quasi-compact universal effective epimorphisms in ZFC (Problem 13). Six of his expectations turned out wrong (Problems 1, 2,

8, 10, 11, and his doubt in Problem 13 about full faithfulness); his programme of classifying topoi, his main conjecture and his doubt about effectivity were borne out.

**Use of language models.** The questions were located in the transcripts, and the literature searched, with language models (Claude Opus 5.5, Anthropic), which also wrote a first draft of this text and the arguments marked [(e)], under the author's direction. The references were opened on their pages; the arguments marked [(e)] have not been checked by a specialist.

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